AN OPEN AND CLOSED MAP WITH CONNECTED TARGET AND ONE FINITE FIBRE HAS AT MOST THAT MANY CONNECTED COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of a connected component of a topological space, and the fact that a space with at least N components decomposes as a disjoint union of N nonempty open and closed pieces. Set up in the author's section on connected components, immediately before the counted statement. Supplies the decomposition that the proof applies to the source of the map.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of nonempty topological spaces. Assume that (a) Y is connected, (b) f is open and closed, and (c) there is a point $y \in Y$ such that the fiber $f^{-1}(y)$ is a finite set. Then X has at most $|f^{-1}(y)|$ connected components. Hence any connected component T of X is open and closed, and $f(T)$ is a nonempty open and closed subset of Y, which is therefore equal to Y.*

Proof. If the topological space X has at least N connected components for some $N \in \mathbf{N}$, we find by induction a decomposition $X = X_1 \amalg \ldots \amalg X_N$ of X as a disjoint union of N nonempty open and closed subsets $X_1, \ldots, X_N$ of X. As f is open and closed, each $f(X_i)$ is a nonempty open and closed subset of Y and is hence equal to Y. In particular the intersection $X_i \cap f^{-1}(y)$ is nonempty for each $1 \leq i \leq N$. Hence $f^{-1}(y)$ has at least N elements. □

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